



Lecture 2: What Is the Object of Assurance?

CS120: Introduction to AI Safety - September 24, 2026, Zachary Robertson

The classifier performs well. Is AI safe?

Safety claims require specifying the intended system under consideration.

The toxicity classifier performs well on held-out labels.

Account-suspension appeals have increased.

What does the evaluation establish - and about what?

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Object, boundary, and unit of analysis

Object of assurance: the bounded artifact, system, or institution about which a claim is actually made

System boundary: specifies the variables intended to be modeled in a system

Unit of analysis: provides a representation for generating well-specified claims

Choose a boundary for the claim

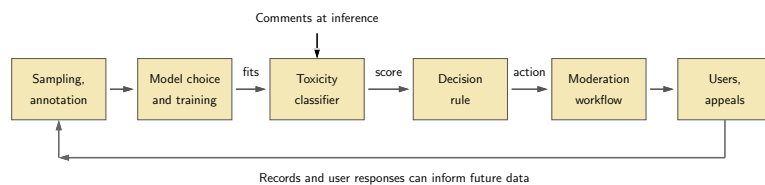
Choose a system boundary narrow enough to analyze, but broad enough to contain the variables and interactions relevant to the claim.

System boundaries are specifications - not inherent factorizations of the world.

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Running example: a content-filter system



Potential hazard: a spurious correlation between dialect and toxicity

However, "where" is the relevant safety problem located?

Models as objects of assurance

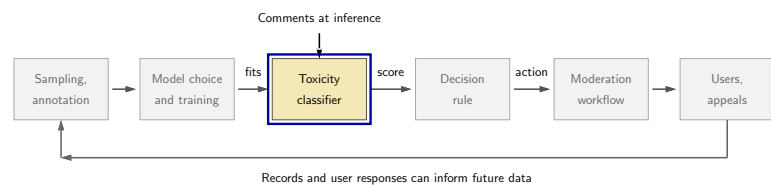


Figure: object of assurance; gray components are environment not irrelevant.

$$f : \mathcal{X} \rightarrow \mathcal{Y}$$

A comment $x \in \mathcal{X}$ is mapped to a toxicity score $y \in \mathcal{Y}$.

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What useful assurances remain?

Question	Required reference
How do false-positive rates compare across linguistic groups?	Specified labels, groups, samples, and a score threshold
Is the score calibrated?	A specified outcome and evaluation distribution
Is there evidence of overfitting?	Information relating fitting performance to held-out performance

What useful assurances remain within this boundary?

Access is not the same as scope

Black-box access

Inputs and outputs

White-box access

Parameters, architecture, or internal activations

Either way, such an object boundary scopes assurance claims.

Development pipelines as objects of assurance

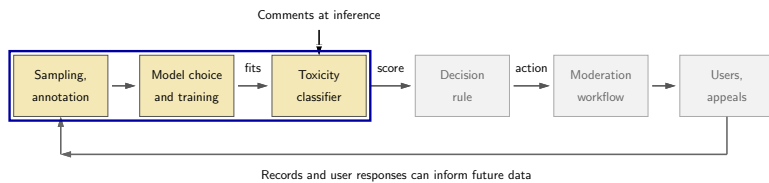
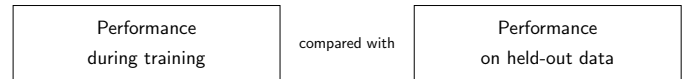


Figure: object of assurance; gray components are environment not irrelevant.

Include data collection, annotation, model and objective selection, and training.

An observed dialect correlation may originate from several places.

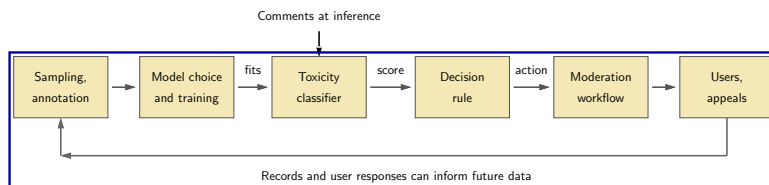
Generalization is not deployment assurance



Generalization does not yet imply good performance under realistic deployment conditions.

Assurance of the model, the process that produced it, and the performance in the intended setting remain distinct problems.

Deployed systems as objects of assurance

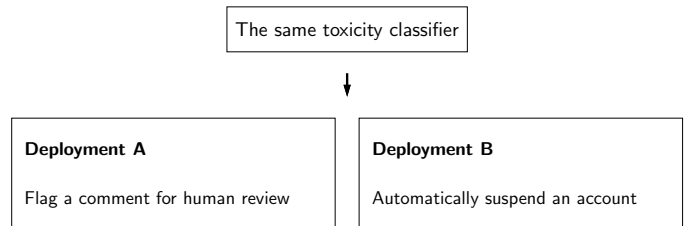


Here the boundary includes relevant user responses and retraining.

The content filter is no longer an isolated scoring function or a well-fit classifier.

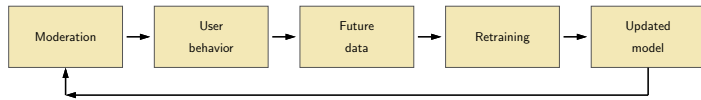
Now its predictions are used for decision-making.

Same classifier, different consequences



Whether a classifier's false-positive rate causes harm depends on the system the model is embedded within.

Deployment creates feedback



Sociotechnical system: technical components interacting with people, social context, and organizational procedures

These interactions often vanish when the technical artifact is abstracted from its original context.

Selbst et al., 2019

The role of institutions

How are safety judgments produced and legitimized?

Who chooses the evaluation criteria?

Are incidents documented, and is the organization accountable for responding?

Who has the authority to pause or remediate a deployment?

Broader is not automatically better

Just because institutions provide coverage of scope does not mean they should always be included in the system boundary.

Claims about an institution cannot be justified solely based on technical artifacts it oversees.

Boundary misspecification

The classifier predicts held-out toxicity labels well.

Does not, by itself, establish

The moderation system is safe.

Evidence concerns one object of assurance while the claim is about another.

Two readings, complementary questions

Concrete Problems in AI Safety

What concrete safety problem can arise in learning or acting?

Amodei et al., 2016

Fairness and Abstraction in Sociotechnical Systems

What relevant interactions vanish in the abstraction?

Selbst et al., 2019

What must be included - or explicitly assumed - to investigate the intended claim?

Match the object, claim, and evidence

Object of assurance

Trained model

Development pipeline

Deployed system

Institution

Focus in the content-filter example

Scoring behavior under specified evaluation conditions

Sampling, annotation, objectives, and training

Moderation decisions, consequences, appeals, and feedback

Evaluation criteria, accountability, and authority to intervene

What exactly are we claiming about it, and what evidence would justify that claim?

Discussion: connect, apply, revise

Start from your reading response; test and refine the connection with classmates.

Questions 1-2: connect the readings: Use a specific idea or example from each paper to explain how an abstraction trap complicates, challenges, or extends the safety problem.

Questions 3-4: apply the comparison: Return to the content filter: choose an object and boundary, then identify evidence and its limit.

Question 5: revise individually: Reconsider your boundary, make an assumption explicit, or narrow your conclusion.

Short answers on the ticket; this is separate from the pre-class reading response.

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Discussion: a concrete safety problem

Amodei et al., 2016

Choose one problem discussed in the paper. What system is being considered, and what could go wrong?

Which assumptions about the objective or environment matter to that problem?

Ground your interpretation in an example or passage from the paper.

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Discussion: an abstraction trap

Selbst et al., 2019

Choose one abstraction trap. In an example from the paper, what relevant relationship is omitted or misrepresented?

Would widening the system boundary address the problem, or would the problem definition itself need to change?

Use the paper's example before comparing it with another system.

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Discussion: return to the content filter

A platform uses a toxicity classifier trained on crowd-sourced labels to trigger automatic account suspensions. Account-suspension appeals have increased. The classifier performs well on held-out toxicity labels.

Distinguish observations from assumptions.

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Choose an object and boundary

Choose one question you would investigate about the increase in appeals.

What is the object of your investigation? What belongs inside the boundary, and what remains an assumption about the environment?

Choose a question narrow enough to investigate.

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Identify evidence - and its limit

What observation would help answer your question? What would it still leave unresolved?

Identify one assumption or omitted relationship that either paper would prompt you to examine.

Separate a useful measurement from the broader claim it might be used to support.

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Revise the boundary or narrow the claim

After discussion, would you change your boundary, make an assumption explicit, or narrow your intended conclusion? Explain why.

Record your own final response. A justified clarification is sufficient.

Next meeting: what counts as an assurance claim?

Specifying the object makes the question precise. It does not yet justify the answer.

[Meeting 3](#): system boundaries, evidence, uncertainty, and systematic hazard analysis

Course website and Ed: readings and response instructions